

Next Generation Data Centers and Energy Management Systems

Lecture 6A: Chip Cooling: Heat Conduction and Thermal Interface Materials

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Learning Plan

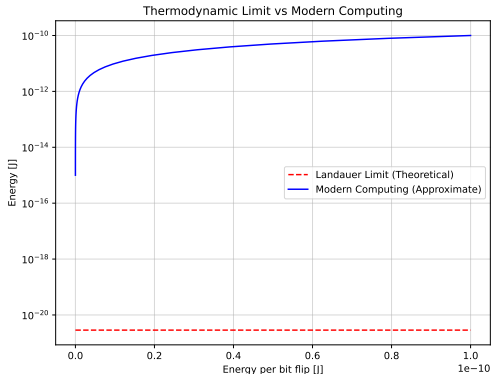
- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- **Chip Cooling: Heat Conduction and Thermal Interface Materials**
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Understand the thermodynamic limits of computing (Landauer's Principle).
- Analyze modern chip architectures and their thermal implications.
- Model heat conduction through a multi-layer chip cooling stack.

Thermodynamics of Computing

- **Why is heat generated?** Irreversible computing operations (e.g., erasing a bit) increase entropy.
- **Landauer's Principle:** The minimum energy required to erase one bit of information is $E_{min} = k_B T \ln 2$.
- **Metric:** Joules per FLOP (or bit flip).
- **Modern vs. Limit:** Modern systems are orders of magnitude away from the thermodynamic limit.



Thermodynamics of Computing Notebook

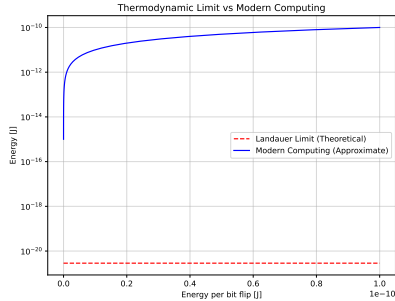


Figure 2: Energy scales in computing.

Modern Chip Architecture

- **Components:**

- Logic Die (CPU/GPU cores)
- Memory (HBM - High Bandwidth Memory)
- Interposer (Silicon or organic substrate connecting die and memory)
- Vias (TSVs - Through-Silicon Vias)

- **Heat Generation:** Concentrated in the logic die (high power density).

- **Thermal Management Stack:**

- Thermal Interface Material (TIM)
- Heat Spreader (Integrated Heat Spreader - IHS)
- Vapor Chamber (Phase change cooling)
- Heat Sink (Convection to air/liquid)

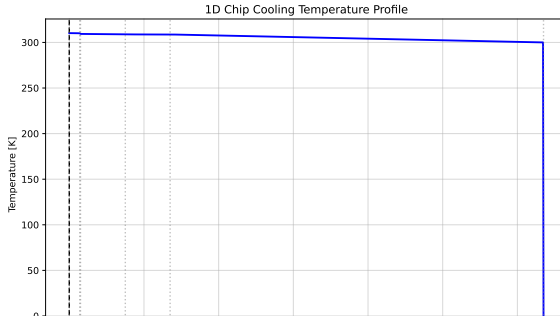
Heat Transfer Modeling

- **Physics:** 1D Steady-state heat conduction with heat generation $\dot{q}$:

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) + \dot{q} = 0$$

- **Application:** Modeling the temperature gradient across the cooling stack:

Silicon $\rightarrow$ TIM $\rightarrow$ Spreader $\rightarrow$ Vapor Chamber $\rightarrow$ Heat Sink



Chip Heat Transfer Notebook

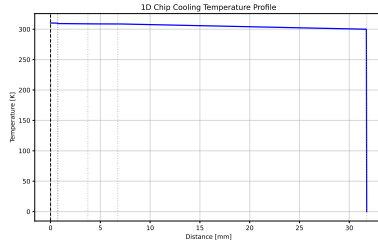


Figure 4: Thermal profile simulation.

Seminar 3

- Name
- Title: TBD
- Date: TBD
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Typical layout of a data center facility

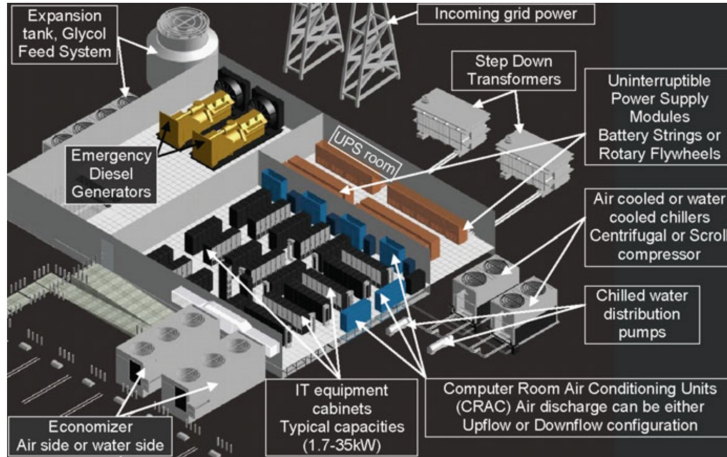


Figure 5: (Figure 1.1 in Ref. [1]).

References I

- [1] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.